

Organic Chemistry: Grand NEET Mock Test

Comprehensive 90-Question Assessment (Final Verified Edition)

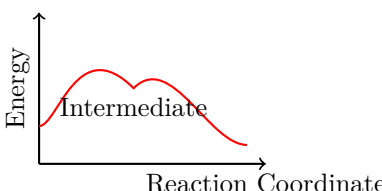
Instructions

- This paper contains **90 multiple-choice questions**.
- Includes **15 integrated Advanced Multi-Step Conversions** (Q76-Q90) matching the latest high-yield patterns.
- Section composition spans Basic Concepts, Hydrocarbons, Haloalkanes, Alcohols, Aldehydes, Amines, and Carbohydrates.
- Marking Scheme: +4 for correct, -1 for incorrect.

Questions

1. The IUPAC name of $\text{CH}_3 - \text{CH} = \text{CH} - \text{COOH}$ is:
 - (a) But-2-enoic acid
 - (b) But-3-enoic acid
 - (c) Prop-2-enoic acid
 - (d) Pent-2-enoic acid
2. The total number of structural isomers possible for the molecular formula $\text{C}_4\text{H}_{10}\text{O}$ is:
 - (a) 4
 - (b) 5
 - (c) 7
 - (d) 8
3. Kjeldahl's method cannot be used for the estimation of nitrogen in:
 - (a) Nitrobenzene
 - (b) Urea
 - (c) Aniline
 - (d) Benzylamine
4. In Lassaigne's test, when both nitrogen and sulphur are present in the organic compound, the blood red colour formed is due to:
 - (a) $\text{Fe}_4[\text{Fe}(\text{CN})_6]_3$
 - (b) $[\text{Fe}(\text{SCN})]^{2+}$
 - (c) $\text{Na}_2[\text{Fe}(\text{CN})_5\text{NO}]$
 - (d) $\text{Fe}(\text{SCN})_3$
5. Paper chromatography is an example of:
 - (a) Partition chromatography
 - (b) Adsorption chromatography
 - (c) Column chromatography
 - (d) Thin layer chromatography
6. Which of the following carbocations is the most stable?
 - (a) $\text{CH}_3 - \text{CH}_2^+$
 - (b) $\text{CH}_3 - \text{C}^+\text{H} - \text{CH}_3$
 - (c) $\text{CH}_2 = \text{CH} - \text{C}^+\text{H}_2$
 - (d) $(\text{CH}_3)_3\text{C}^+$
7. Which of the following acts as an electrophile?
 - (a) NH_3
 - (b) H_2O
 - (c) BF_3
 - (d) CH_3NH_2
8. Which of the following acts as a nucleophile?
 - (a) AlCl_3
 - (b) CN^-
 - (c) NO_2^+
 - (d) SO_3
9. The number of hyperconjugation structures possible for propene ($\text{CH}_3 - \text{CH} = \text{CH}_2$) is:
 - (a) 2
 - (b) 3
 - (c) 4
 - (d) 5
10. In Dumas' method for estimation of nitrogen, the gas collected in the nitrometer is:
 - (a) N_2
 - (b) NO
 - (c) NO_2
 - (d) NH_3
11. Tautomerism is exhibited by:
 - (a) Benzophenone
 - (b) Benzaldehyde
 - (c) Acetaldehyde
 - (d) Diethyl ether
12. Metamerism is exhibited by which of the following pairs?
 - (a) Diethyl ether and Methyl propyl ether
 - (b) Ethanol and Dimethyl ether
 - (c) Butane and Isobutane
 - (d) Propan-1-ol and Propan-2-ol

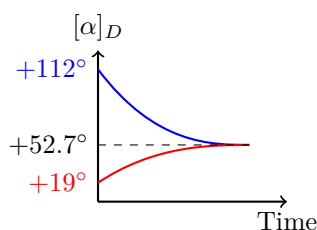
13. Geometrical isomerism is strictly shown by:
- But-1-ene
 - But-2-ene
 - Propene
 - Ethene
14. A meso compound is optically inactive due to:
- Internal compensation
 - External compensation
 - Absence of chiral centers
 - Steric hindrance
15. According to Hückel's rule, a cyclic, planar, conjugated system is aromatic if it contains:
- $4n\pi$ electrons
 - $(2n + 2)\pi$ electrons
 - $(4n + 2)\pi$ electrons
 - $4n + 1$ electrons
16. Which of the following is an ortho-para directing group in electrophilic aromatic substitution?
- $-\text{OH}$
 - $-\text{NO}_2$
 - $-\text{COOH}$
 - $-\text{CHO}$
17. Which of the following is a meta directing group?
- $-\text{CH}_3$
 - $-\text{NH}_2$
 - $-\text{OCH}_3$
 - $-\text{NO}_2$
18. The most acidic compound among the following is:
- Phenol
 - p-Nitrophenol
 - p-Cresol
 - m-Cresol
19. The correct order of electron-withdrawing effect (-I effect) is:
- $-\text{NO}_2 > -\text{F} > -\text{OH}$
 - $-\text{F} > -\text{NO}_2 > -\text{OH}$
 - $-\text{OH} > -\text{F} > -\text{NO}_2$
 - $-\text{NO}_2 > -\text{OH} > -\text{F}$
20. The stability order of alkyl free radicals is:
- $3^\circ > 2^\circ > 1^\circ$
 - $1^\circ > 2^\circ > 3^\circ$
 - $2^\circ > 3^\circ > 1^\circ$
 - $3^\circ = 2^\circ = 1^\circ$
21. Kolbe's electrolytic method applied to potassium succinate yields:
- Ethane
 - Ethene
 - Ethyne
 - Methane
22. Reductive ozonolysis of But-2-ene yields:
- Formaldehyde
 - Acetone
 - Acetaldehyde
 - Formic acid
23. Markownikoff's rule determines the major product in the addition of HBr to:
- Propene
 - But-2-ene
 - Ethene
 - Ethyne (symmetrical)
24. The anti-Markownikoff addition (Peroxide effect) is observed only with:
- HCl
 - HBr
 - HI
 - HF
25. Baeyer's reagent, used as a test for unsaturation, is:
- Cold, dilute, alkaline KMnO_4
 - Hot, concentrated KMnO_4
 - Acidic $\text{K}_2\text{Cr}_2\text{O}_7$
 - Bromine water
26. Hydration of ethyne in the presence of Hg^{2+} and H_2SO_4 gives:
- Ethanol
 - Ethanoic acid
 - Acetaldehyde
 - Acetone
27. Partial reduction of alkynes using Lindlar's catalyst ($\text{H}_2/\text{Pd} - \text{BaSO}_4$) yields:
- cis-Alkenes
 - trans-Alkenes
 - Alkanes
 - Alcohols
28. Birch reduction of internal alkynes yields:
- cis-Alkenes
 - trans-Alkenes
 - Alkanes

- (d) Polymers
29. Benzene reacts with excess Cl_2 in the presence of sunlight to yield:
- Chlorobenzene
 - Dichlorobenzene
 - BHC (Gammexane)
 - Toluene
30. Vigorous oxidation of toluene with acidic KMnO_4 yields:
- Benzaldehyde
 - Benzoic acid
 - Benzyl alcohol
 - Phenol
31. The given potential energy profile corresponds to which of the following reaction mechanisms?
- 
32. Walden inversion is a characteristic feature of:
- $\text{S}_{\text{N}}2$ reactions
 - $\text{S}_{\text{N}}1$ reactions
 - $\text{E}1$ reactions
 - $\text{E}2$ reactions
33. The correct reactivity order of alkyl halides towards $\text{S}_{\text{N}}1$ reactions is:
- $3^\circ > 2^\circ > 1^\circ$
 - $1^\circ > 2^\circ > 3^\circ$
 - $2^\circ > 3^\circ > 1^\circ$
 - $1^\circ > 3^\circ > 2^\circ$
34. The correct reactivity order of alkyl halides towards $\text{S}_{\text{N}}2$ reactions is:
- $1^\circ > 2^\circ > 3^\circ$
 - $3^\circ > 2^\circ > 1^\circ$
 - $2^\circ > 1^\circ > 3^\circ$
 - $3^\circ > 1^\circ > 2^\circ$
35. Dehydrohalogenation of an alkyl halide follows:
- Saytzeff's rule
 - Markownikoff's rule
 - Hofmann's rule
 - Hund's rule
36. The Finkelstein reaction involves the conversion of:
- $\text{R}-\text{OH}$ to $\text{R}-\text{Cl}$
 - $\text{R}-\text{H}$ to $\text{R}-\text{Br}$
 - $\text{R}-\text{Cl}$ to $\text{R}-\text{I}$ using $\text{NaI}/\text{acetone}$
 - $\text{R}-\text{Br}$ to $\text{R}-\text{F}$ using AgF
37. The Dow process is used for the commercial preparation of:
- Chlorobenzene
 - Benzoic acid
 - Phenol
 - Aniline
38. Grignard reagents (RMgX) must be prepared in the presence of:
- Dry ether
 - Aqueous ethanol
 - Dilute HCl
 - Liquid ammonia
39. The Wurtz-Fittig reaction between chlorobenzene and methyl chloride in presence of sodium yields:
- Toluene
 - Biphenyl
 - Ethane
 - Benzene
40. Freon-12, a widely known CFC, has the chemical formula:
- CF_4
 - CCl_2F_2
 - C_2F_6
 - CHCl_3
41. The Carbylamine reaction is given by:
- Primary amines only
 - Secondary amines only
 - Tertiary amines only
 - All classes of amines
42. Lucas reagent is a mixture of:
- Conc. HNO_3 and Conc. H_2SO_4
 - Pd/BaSO_4 and quinoline
 - Anhydrous ZnCl_2 and Conc. HCl
 - SnCl_2 and HCl
43. Hydroboration-oxidation of propene yields:
- Propan-2-ol
 - Propan-1-ol

- (c) Propane
(d) Propanal
44. Oxymercuration-demercuration of propene yields:
(a) Propan-2-ol
(b) Propan-1-ol
(c) Propane
(d) Propanal
45. Reimer-Tiemann reaction of phenol with chloroform and aqueous NaOH produces:
(a) Salicylaldehyde
(b) Salicylic acid
(c) Benzaldehyde
(d) Benzoic acid
46. Kolbe's reaction of phenol produces:
(a) Salicylaldehyde
(b) Salicylic acid
(c) Phenyl acetate
(d) Picric acid
47. Heating phenol with zinc dust yields:
(a) Benzene
(b) Toluene
(c) Cyclohexane
(d) Benzyl alcohol
48. Williamson synthesis of ethers proceeds via which mechanism?
(a) S_N1
(b) S_N2
(c) E1
(d) E2
49. Cleavage of anisole ($C_6H_5 - O - CH_3$) with concentrated HI yields:
(a) Phenol + Methyl iodide
(b) Iodobenzene + Methanol
(c) Benzene + Methanol
(d) Phenol + Methane
50. Dehydration of ethanol with concentrated H_2SO_4 at 443 K yields:
(a) Ethene
(b) Diethyl ether
(c) Ethane
(d) Ethanal
51. Dehydration of ethanol with concentrated H_2SO_4 at 413 K predominantly yields:
(a) Ethene
(b) Diethyl ether
(c) Ethane
(d) Ethanal
52. Aspirin is synthesized by the acetylation of:
(a) Salicylic acid
(b) Salicylaldehyde
(c) Phenol
(d) Benzoic acid
53. The reagent used in the Etard reaction to oxidize toluene to benzaldehyde is:
(a) $KMnO_4$
(b) CrO_2Cl_2 (Chromyl chloride)
(c) CrO_3 with acetic anhydride
(d) $K_2Cr_2O_7$
54. Gattermann-Koch reaction uses CO and HCl in presence of anhydrous $AlCl_3/CuCl$ to form:
(a) Benzaldehyde
(b) Acetophenone
(c) Benzoic acid
(d) Benzophenone
55. Aldol condensation is exhibited by aldehydes and ketones containing at least:
(a) One α -hydrogen atom
(b) One β -hydrogen atom
(c) One α -OH group
(d) No α -hydrogen atoms
56. Cannizzaro reaction is a disproportionation reaction strictly given by:
(a) Acetaldehyde
(b) Acetone
(c) Formaldehyde
(d) Propionaldehyde
57. Clemmensen reduction of carbonyl compounds uses:
(a) Zn-Hg / conc. HCl
(b) NH_2NH_2 / KOH
(c) $LiAlH_4$
(d) H_2 / Pd
58. Wolff-Kishner reduction reduces carbonyl compounds to alkanes using:
(a) Zn-Hg / conc. HCl
(b) NH_2NH_2 / KOH / Ethylene glycol
(c) $NaBH_4$
(d) Red P / HI
59. Iodoform test is positive for compounds containing the group:

- (a) $\text{CH}_3 - \text{CO}-$
 (b) $\text{CH}_3 - \text{CH}_2 - \text{CO}-$
 (c) $-\text{CHO}$ (only if lacking alpha methyl)
 (d) $-\text{COOH}$
- 60.** Tollens' reagent reacts with aldehydes to form a silver mirror. The reagent is:
 (a) Ammoniacal silver nitrate
 (b) Alkaline copper sulphate
 (c) Acidic potassium dichromate
 (d) Sodium nitroprusside
- 61.** The Hell-Volhard-Zelinsky (HVZ) reaction halogenates carboxylic acids specifically at the:
 (a) Carboxyl carbon
 (b) α -position
 (c) β -position
 (d) Terminal position
- 62.** Esterification of a carboxylic acid with an alcohol is mechanistically an example of:
 (a) Nucleophilic acyl substitution
 (b) Electrophilic addition
 (c) Free radical substitution
 (d) Elimination
- 63.** The base-catalyzed hydrolysis of an ester is termed:
 (a) Hydration
 (b) Transesterification
 (c) Saponification
 (d) Decarboxylation
- 64.** Gabriel phthalimide synthesis is exclusively used for the preparation of:
 (a) Primary aliphatic amines
 (b) Secondary amines
 (c) Tertiary amines
 (d) Primary aromatic amines
- 65.** Hoffmann bromamide degradation converts an amide into a primary amine containing:
 (a) One carbon more than the amide
 (b) One carbon less than the amide
 (c) The exact same number of carbons
 (d) Two carbons less than the amide
- 66.** The order of basic strength of methyl-substituted amines in aqueous solution is:
 (a) $2^\circ > 1^\circ > 3^\circ > \text{NH}_3$
 (b) $1^\circ > 2^\circ > 3^\circ > \text{NH}_3$
 (c) $3^\circ > 2^\circ > 1^\circ > \text{NH}_3$
 (d) $\text{NH}_3 > 1^\circ > 2^\circ > 3^\circ$
- 67.** The order of basic strength of ethyl-substituted amines in aqueous solution is:
 (a) $2^\circ > 1^\circ > 3^\circ > \text{NH}_3$
 (b) $1^\circ > 2^\circ > 3^\circ > \text{NH}_3$
 (c) $2^\circ > 3^\circ > 1^\circ > \text{NH}_3$
 (d) $3^\circ > 2^\circ > 1^\circ > \text{NH}_3$
- 68.** Hinsberg's reagent, used to distinguish primary, secondary, and tertiary amines, is:
 (a) Benzenesulfonyl chloride
 (b) Benzoyl chloride
 (c) Acetyl chloride
 (d) p-Toluenesulfonyl chloride
- 69. Statement I:** Aniline undergoes Friedel-Crafts alkylation smoothly.
Statement II: Aniline forms a salt with the Lewis acid AlCl_3 .
 (a) Statement I is incorrect, Statement II is correct.
 (b) Both are correct.
 (c) Statement I is correct, Statement II is incorrect.
 (d) Both are incorrect.
- 70.** Reaction of a primary aliphatic amine with nitrous acid (HNO_2) primarily yields:
 (a) A stable diazonium salt
 (b) Secondary amine
 (c) Alcohol and N_2 gas
 (d) Nitroalkane
- 71.** Naturally occurring glucose is structurally:
 (a) D-(+)-Glucose
 (b) L-(-)-Glucose
 (c) D-(-)-Glucose
 (d) L-(+)-Glucose
- 72.** Oxidation of glucose with mild oxidizing agent like bromine water yields:
 (a) Saccharic acid
 (b) Gluconic acid
 (c) Sorbitol
 (d) Fructose
- 73.** Oxidation of glucose with strong oxidizing agent like nitric acid yields:
 (a) Saccharic acid
 (b) Gluconic acid
 (c) Glucuronic acid only
 (d) Glyceraldehyde

74. The given graph represents the change in specific rotation over time. This phenomenon is known as:



- (a) Inversion
(b) Racemization
(c) Mutarotation
(d) Epimerization

75. Match the Column:

Column I

- (A) Rosenmund Reduction
(B) Clemmensen Reduction
(C) Wolff-Kishner Reduction
(D) Etard Reaction

Column II

- (p) $\text{NH}_2\text{NH}_2/\text{KOH}$
(q) $\text{H}_2/\text{Pd} - \text{BaSO}_4$
(r) CrO_2Cl_2
(s) $\text{Zn} - \text{Hg}/\text{HCl}$

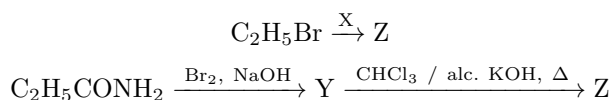
- (a) A-q, B-s, C-p, D-r
(b) A-s, B-q, C-r, D-p
(c) A-q, B-p, C-r, D-s
(d) A-r, B-p, C-q, D-s

76. In the following sequence of reactions, identify the final product **Z**:



- (a) Propanoic acid
(b) Ethanoic acid
(c) Propanal
(d) Propan-1-amine

77. The following two reactions give the same foul-smelling product **Z**.



Identify **X** and **Y** respectively:

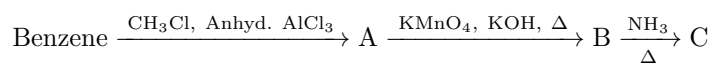
- (a) **X** = KCN; **Y** = $\text{C}_2\text{H}_5\text{NH}_2$
(b) **X** = AgCN; **Y** = $\text{C}_2\text{H}_5\text{NH}_2$
(c) **X** = AgCN; **Y** = $\text{C}_2\text{H}_5\text{CH}_2\text{NH}_2$
(d) **X** = KCN; **Y** = CH_3NH_2

78. Compound **A** ($\text{C}_3\text{H}_6\text{O}$) does not reduce Fehling's solution but forms a 2,4-DNP derivative. It reacts with I_2 and NaOH to give a yellow precipitate of **B**. What is the structure of **A** and **B**?

- (a) **A** = Propanal; **B** = Iodoform
(b) **A** = Propanone; **B** = Iodoethane

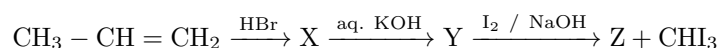
- (c) **A** = Propanone; **B** = CHI_3
(d) **A** = Propanal; **B** = CHI_3

79. In the following reaction sequence, identify the structure of **C**:



- (a) Toluene
(b) Benzoic acid
(c) Benzamide
(d) Aniline

80. Consider the following sequence of reactions:



Identify **Z**:

- (a) CH_3COONa
(b) $\text{CH}_3\text{CH}_2\text{COONa}$
(c) HCOONa
(d) CH_3CHO

81. Phenol reacts with **X** in the presence of NaOH to form Salicylaldehyde. This reaction is known as the Reimer-Tiemann reaction. Identify **X** and the reaction intermediate **Y**:

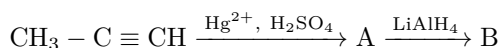
- (a) **X** = CCl_4 ; **Y** = Carbocation
(b) **X** = CHCl_3 ; **Y** = Dichlorocarbene ($:\text{CCl}_2$)
(c) **X** = CO_2 ; **Y** = Carbanion
(d) **X** = CH_2Cl_2 ; **Y** = Free radical

82. Identify **D** in the following sequence:



- (a) Benzene
(b) Benzyl alcohol
(c) Phenol
(d) Benzoic acid

83. Predict the major product **B** in the following reaction sequence:



- (a) Propan-1-ol
(b) Propan-2-ol
(c) Propane
(d) Propanal

84. An organic compound **P** ($\text{C}_2\text{H}_4\text{O}$) reduces Tollens' reagent. **P** on treatment with dilute NaOH undergoes a reaction to form **Q**. **Q** on heating eliminates a water molecule to form **R**. Identify **R**.

- (a) But-2-enal
- (b) Butan-1-ol
- (c) But-2-enoic acid
- (d) Ethoxyethane

85. Identify the final product **C** in the given reaction:

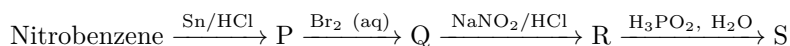


- (a) Propanoyl chloride
- (b) 2-Oxopropanoic acid (Pyruvic acid)
- (c) Propan-2-ol
- (d) Lactic acid

86. Compound **A** reacts with PCl_5 to give **B**. Compound **B** reacts with alcoholic KOH to give **C**. Ozonolysis of **C** gives only Formaldehyde (HCHO). What is **A**?

- (a) Methanol
- (b) Ethanol
- (c) Propan-1-ol
- (d) Propan-2-ol

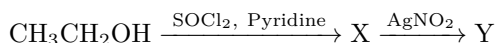
87. Consider the transformation:



The structure of **S** is:

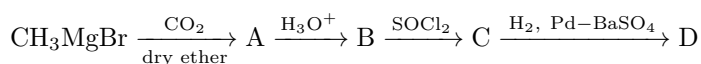
- (a) Benzene
- (b) 1,3,5-Tribromobenzene
- (c) 2,4,6-Tribromophenol
- (d) p-Bromonitrobenzene

88. In the following sequence, what are **X** and **Y**?



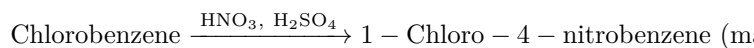
- (a) **X** = $\text{CH}_3\text{CH}_2\text{Cl}$; **Y** = Nitroethane
- (b) **X** = $\text{CH}_3\text{CH}_2\text{Cl}$; **Y** = Ethyl nitrite
- (c) **X** = $\text{CH}_2 = \text{CH}_2$; **Y** = $\text{NO}_2\text{CH}_2\text{CH}_2\text{OH}$
- (d) **X** = CH_3CHO ; **Y** = CH_3COOH

89. Identify **D** in the reaction:



- (a) Ethanoic acid
- (b) Ethanal
- (c) Ethanol
- (d) Acetone

90. Identify the reagent **R** and the product **Y** in the following reaction sequence:



- (a) **R** = aq. NaOH at 443 K; **Y** = p-Nitrophenol
- (b) **R** = aq. NaOH at 623 K; **Y** = Phenol
- (c) **R** = Sn/HCl ; **Y** = p-Chloroaniline
- (d) **R** = CH_3Cl , Anhyd. AlCl_3 ; **Y** = 2-Chloro-4-nitrotol

Answer Key

1. (a)	19. (a)	37. (c)	55. (a)	73. (a)
2. (c)	20. (a)	38. (a)	56. (c)	74. (c)
3. (a)	21. (b)	39. (a)	57. (a)	75. (a)
4. (b)	22. (c)	40. (b)	58. (b)	76. (a)
5. (a)	23. (a)	41. (a)	59. (a)	77. (b)
6. (d)	24. (b)	42. (c)	60. (a)	78. (c)
7. (c)	25. (a)	43. (b)	61. (b)	79. (c)
8. (b)	26. (c)	44. (a)	62. (a)	80. (a)
9. (b)	27. (a)	45. (a)	63. (c)	81. (b)
10. (a)	28. (b)	46. (b)	64. (a)	82. (b)
11. (c)	29. (c)	47. (a)	65. (b)	83. (b)
12. (a)	30. (b)	48. (b)	66. (a)	84. (a)
13. (b)	31. (b)	49. (a)	67. (c)	85. (b)
14. (a)	32. (a)	50. (a)	68. (a)	86. (b)
15. (c)	33. (a)	51. (b)	69. (a)	87. (b)
16. (a)	34. (a)	52. (a)	70. (c)	88. (a)
17. (d)	35. (a)	53. (b)	71. (a)	89. (b)
18. (b)	36. (c)	54. (a)	72. (b)	90. (a)